

ENGINEERING GRAPHICS

MODULE-WISE MOST IMPORTANT TOPICS (PYQs BASED)

MODULE-1: Scales & Engineering Curves

Top 5 Most Important Topics

1. **Diagonal Scale (construction + numericals)**

Asked in: 2019, 2020, Jun 2022, Nov 2022, Jun 2023, Jun 2024, Jun 2025

2. **Vernier Scale (forward / backward)**

Asked in: Jun 2022, Jun 2024, Jun 2025

3. **Conic Sections (Ellipse / Parabola / Hyperbola)**

Asked in: 2020, Jun 2022, Nov 2022, Dec 2024, Jun 2025

4. **Cycloid / Epicycloid / Hypocycloid / Involute**

Asked in: 2019, Jun 2022, Jun 2023, Jun 2025

5. **RF, Types & Uses of Scales (theory)**

Asked in: 2019, Jun 2023, Jun 2024

Conclusion: *Most scoring + most repeated module*

MODULE-2: Orthographic Projection & Lines

Top 5 Most Important Topics

1. **Projection of Straight Line (inclined to HP & VP)**

Asked in: 2019, 2020, Jun 2022, Nov 2022, Jun 2023, Dec 2024, Jun 2025

2. **True Length & True Inclination of Line**

Asked in: Jun 2022, Jun 2023, Jun 2024

3. Traces of a Line

Asked in: **2020, Jun 2023, Jun 2025**

4. Projection of Points (quadrants)

Asked in: **Dec 2023, Jun 2023**

5. First Angle vs Third Angle Projection

Asked in: **2020, Jun 2022, Dec 2024, Jun 2025**

MODULE–3: Projection of Planes

Top 5 Most Important Topics

1. Regular Plane Lamina (triangle/square/hexagon)

Asked in: **2019, Jun 2022, Jun 2023, Dec 2024, Jun 2025**

2. Plane inclined to HP & VP (auxiliary plane)

Asked in: **Jun 2023, Jun 2024, Jun 2025**

3. Rectangular / Rhombus Plate problem

Asked in: **Dec 2024, Jun 2024**

4. True Shape of Plane

Asked in: **Jun 2024, Jun 2025**

5. Plane resting on edge / corner

Asked in: **2019, Jun 2023**

MODULE–4: Sections & Section of Solids

Top 5 Most Important Topics

1. Projection of Solids (prism, pyramid, cone, cylinder)

Asked in: **EVERY YEAR (2019–2025)**

2. Axis inclined to HP / VP

Asked in: **Jun 2022, Nov 2022, Jun 2023, Jun 2025**

3. Section of Solids + True Shape of Section

Asked in: **2019, Jun 2023, Jun 2024, Jun 2025**

4. Frustum of Cone / Pyramid

Asked in: **Jun 2022, Jun 2024**

5. Combination of Solid

Asked in: **2019, Jun 2023, Jun 2025**

MODULE–5: Isometric Projection & Development

Top 5 Most Important Topics

1. Isometric Projection / Isometric View

Asked in: **2019, 2020, Jun 2022, Jun 2023, Jun 2024, Jun 2025**

2. Isometric Scale (construction & use)

Asked in: **Jun 2023, Jun 2025**

3. Development of Surfaces (cone, cylinder)

Asked in: **2019, Jun 2023, Jun 2024**

4. Conversion: Orthographic ↔ Isometric

Asked in: **Jun 2024, Jun 2025**

5. Compound Solids (isometric)

Asked in: **2019, Jun 2023**

MODULE–6/7/8: CAD, Drafting & BIM (Theory Oriented)

Top Repeated Topics

1. CAD advantages & applications – 5–7 marks

2019, 2020, Jun 2022, Nov 2022, Jun 2025

2. Editing / Modify commands (Move, Copy, Rotate, Mirror)

2020, Jun 2022, Jun 2024

3. **Layering concept, Viewports, Zoom, UCS**

Jun 2023, Dec 2023, Jun 2025

4. **BIM – concept & uses**

Jun 2022, Dec 2024

5. **Dialog boxes & CAD window**

Jun 2025

PREDICTED MODEL QUESTION PAPER (Set-1)

(Based on PYQs Analysis)

Time: 3 Hours **Maximum Marks:** 70

Instructions:

1. Attempt **any five** questions.
2. All questions carry **equal marks**.
3. Use neat sketches and proper dimensioning.
4. Assume suitable data wherever necessary.

Q1 – SCALES (Very High Probability)

a) Construct a **Diagonal Scale** to read **meters, decimeters and centimeters** to measure up to **6 m**, when **1 m is represented by 2.5 cm**. Mark a distance of **4.28 m** on the scale.

OR

b) What is a **scale**? Classify different types of scales. Define **R.F.** and explain its importance with a neat sketch.

Q2 – ENGINEERING CURVES (Sure Question)

a) Construct an **Ellipse** by **concentric circle method** when major axis = **100 mm** and minor axis = **75 mm**. Draw a **tangent and normal** at a given point.

OR

b) Draw an **Epicycloid** for a generating circle of **50 mm diameter** rolling on a directing circle of **150 mm diameter**.

Q3 – PROJECTION OF LINES (Very Repeated)

a) A line **AB 70 mm long** is inclined at **45° to HP** and **30° to VP**. End **A** is **10 mm above HP** and **15 mm in front of VP**. Draw its projections and find the **true length**.

OR

b) Explain **First Angle Projection** and **Third Angle Projection** with neat sketches. Also state their applications.

Q4 – PROJECTION OF PLANES (Guaranteed Drawing)

a) A **regular hexagonal lamina of side 30 mm** rests on HP on one of its sides. The surface is inclined at **45° to HP** and the resting side makes **30° with VP**. Draw its projections.

OR

b) A **rectangular plate 45 mm × 35 mm** has its diagonal inclined at **30° to HP** and the other diagonal makes **45° with VP**. Draw its projections.

Q5 – PROJECTION OF SOLIDS (High Weight)

a) A **triangular prism**, base side **30 mm** and axis **55 mm long**, rests on HP on one of its rectangular faces with the axis **parallel to VP**. Draw its projections.

OR

b) A **right circular cone** of base diameter **50 mm** and axis **60 mm long** rests on HP on its base. One of its generators is **perpendicular to HP**. Draw its projections.

Q6 – SECTION OF SOLIDS (14-Mark Type)

a) A **square pyramid**, base **35 mm** and height **50 mm**, rests on HP. A section plane **perpendicular to VP** and inclined at **32° to HP** cuts the pyramid **bisecting its axis**. Draw the projections and obtain the **true shape of section**.

OR

b) A **right circular cylinder**, base diameter **50 mm** and height **65 mm**, rests on HP. It is cut by a plane inclined at **45° to HP**. Draw the **sectional top view** and **true shape of section**.

Q7 – ISOMETRIC / DEVELOPMENT (Sure Drawing)

a) Draw the **Isometric Projection** of a **cube of 30 mm edge** placed centrally on the top of a **cylindrical block of 50 mm diameter and 20 mm height**.

OR

b) Draw the **development of the lateral surface** of a **right circular cone** of base diameter **40 mm** and height **40 mm**.

Q8 – CAD / THEORY (Easy Marks)

Write **short notes** on any **TWO** of the following:

- a) **Editing / Modify commands in CAD**
 - b) **Layering concept in CAD**
 - c) **Purpose of Zoom command**
 - d) **Advantages and applications of CAD**
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PREDICTED MODEL QUESTION PAPER (Set-2)

Time: 3 Hours

Maximum Marks: 70

Instructions:

1. Attempt **any five** questions.
 2. All questions carry **equal marks**.
 3. Neat diagrams and proper dimensioning are compulsory.
 4. Assume suitable dimensions wherever required.
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Q1 – SCALES (Sure-Shot Question)

- a) The distance between two cities is **100 km** and it is represented on a map by **25 cm**.
- i) Find the **R.F.**
- ii) Construct a **Diagonal Scale** to read up to **400 km** and show a distance of **275 km**.

OR

- b) Define **Vernier Scale**. Construct a **forward reading Vernier scale** to read up to **6 m** when **1 m = 2 cm** and mark **3.65 m** on it.
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Q2 – ENGINEERING CURVES (High Probability)

- a) Construct a **Parabola** when the distance between **focus and directrix is 60 mm**. Draw a **tangent and normal** at a given point.

OR

- b) Draw a **Hypocycloid**, the rolling circle being **50 mm diameter** and the directing circle **175 mm diameter**.
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Q3 – PROJECTION OF LINES

- a) A line **AB 80 mm long** has its end **A 15 mm above HP** and **20 mm in front of VP**. The line is inclined at **30° to HP** and **45° to VP**. Draw its projections and find **true length**.

OR

- b) What is **Orthographic Projection**? Explain the **principle of projection** and differentiate between **First Angle and Third Angle Projection** with neat sketches.

Q4 – PROJECTION OF PLANES

a) A **square lamina of side 40 mm** rests on HP on one of its sides. The surface is inclined at **45° to HP** and the resting side makes **30° with VP**. Draw its projections.

OR

b) Explain **Auxiliary Plane Method** for finding the true shape of a plane with a neat diagram.

Q5 – PROJECTION OF SOLIDS

a) A **pentagonal prism**, base side **25 mm** and axis **50 mm long**, rests on HP on one of its rectangular faces with its axis **inclined at 45° to HP**. Draw its projections.

OR

b) A **right circular cylinder**, base diameter **40 mm** and height **60 mm**, rests on HP with its axis **inclined at 30° to VP**. Draw its projections.

Q6 – SECTION OF SOLIDS (Very Important)

a) A **hexagonal pyramid**, base side **30 mm** and axis **60 mm long**, rests on HP. It is cut by a **horizontal section plane** at a distance of **35 mm above the base**. Draw the **sectional front view and sectional top view**.

OR

b) A **right circular cone**, base diameter **55 mm** and axis **65 mm**, rests on HP. It is cut by a **vertical section plane parallel to one of the generators**. Draw the **sectional front view and true shape of section**.

Q7 – ISOMETRIC PROJECTION

a) Draw the **Isometric View** of a **hexagonal prism** of base side **25 mm** and axis **65 mm long**, resting on its base on HP.

OR

b) Explain **Isometric Scale**. Draw an **isometric projection of a cube of 40 mm side** using isometric scale.

Q8 – CAD / THEORY (Scoring Section)

Write **short notes on any TWO** of the following:

- a) **Layering concept and its advantages**
 - b) **Editing commands: Move, Copy, Rotate, Mirror**
 - c) **Viewports and their uses**
 - d) **Building Information Modeling (BIM)**
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